

DATA

NEXUS

DataNexus Era 3 is the third generation of data platform. Era 1: on-premises servers. Era 2: rented cloud servers. Era 3: data runs everywhere — phones, satellites, sensors, servers — united by blockchain, governed by its own laws. Built entirely on Apache open-source, 12 founding principles. Zero proprietary dependencies. Zero cloud lock-in. Zero compromise on sovereignty.

MANIFESTO

12 founding principles

Era 3 — Technical Specification

1.1 The Six Innovations

#	Innovation	Description
01	Living Data Nodes	Every device becomes a fabric node — phones, IoT, satellites, hospital machines.
02	Trustless Data Fabric	Peer-to-peer mesh on Hyperledger Fabric — cannot be seized or shut down.
03	Zero Gravity Data	Data identified by SHA-256 hash, not path. Flows to compute automatically.
04	AI Operating System	Human declares intent; AI builds, deploys, monitors, heals — autonomously.
05	Data DNA	Dataset carries its own genome: lineage, quality, consent, laws, access policy.
06	Conscious Compliance	Data refuses illegal border crossings. DPDP/GDPR encoded as smart contracts.

1.2 Technology Stack

Layer	Open-Source Tool	Replaces (Cloud)	DataNexus Adds
Ingest	Kafka + NiFi + Sqoop	Kinesis / Pub-Sub / Event Hubs	Blockchain event stamp
Store	HDFS + Iceberg + IPFS	S3 / GCS / ADLS	Content-hash addressing
Process	Spark + Flink + YARN	EMR / Dataproc / HDInsight	AI self-optimizer
Query	Presto + Hive + Druid	Redshift / BigQuery / Synapse	NLP in any language
Orchestrate	Airflow (self-healing)	MWAA / Composer / ADF	Auto-heal plugin
Govern	Atlas + Ranger + GE	Lake Formation / Purview	Hyperledger audit
Blockchain	Hyperledger Fabric	None — no cloud has this	Tamper-proof lineage
ML/AI	Spark MLlib + MLflow	SageMaker / Vertex AI	Autonomous pipelines
BI	Superset + Druid	Tableau / Power BI	Live sigma overlay

6

Innovations

8

Layers

5

Modules

0

Cloud deps

2. Data DNA — Cryptographic Dataset Genome

Every dataset in DataNexus carries a DataDNA object. The genome_hash field is SHA-256 of all 12 DNA fields. Any tampering changes the hash — instantly detectable. Data carries its own truth forever.

2.1 DNA Fields

Field	Type	Purpose
dataset_id	UUID	Immutable unique identifier
creator_id	str	Organisation or individual who owns this data
classification	Enum	PUBLIC / CONFIDENTIAL / PII / HEALTH / FINANCIAL
jurisdictions	List[Enum]	Applicable laws: DPDP / GDPR / HIPAA / SOX
parent_ids	List[str]	Upstream datasets — complete lineage chain
quality.sigma	float 1-6	Six Sigma quality level from Great Expectations
consent	Record	Owner, purpose, grant time, expiry, revocability
border_policy	Policy	Allowed regions, blocked regions, multi-sig rules
content_hash	SHA-256	Hash of raw data bytes — changes if data tampered
ipfs_cid	CID	IPFS content identifier for distributed storage
fabric_tx_id	str	Hyperledger Fabric TX — permanent ledger proof
genome_hash	SHA-256	Hash of ALL above fields — detects DNA tampering

2.2 Border Autonomy

can_cross_border(target_region) evaluates jurisdiction rules inside the data object. DPDP rule: Indian PII cannot leave region prefix IN without explicit consent + 3 signatures. GDPR rule: EU personal data cannot leave EU without adequacy decision. No external system is queried — the data decides for itself in milliseconds.

```
allowed, reason = dna.can_cross_border("US")
# → False, "BLOCKED: DPDP 2023 – cannot leave India"

allowed, reason = dna.can_cross_border("IN-TG")
# → True, "ALLOWED: IN-TG – requires 3 signatures"

print(dna.verify_integrity()) # True
dna.creator_id = "tampered"
print(dna.verify_integrity()) # False – detected instantly
```

3. Zero Gravity Data Fabric

Data in the Zero Gravity Fabric has no fixed address. It is identified by SHA-256 content hash via IPFS. When a Spark job registers compute gravity on a node, the fabric automatically replicates the needed datasets to that node. Data flows like water — toward where it is needed.

Concept	Traditional Cloud	DataNexus Zero Gravity
Data address	s3://bucket/path/file.parquet	SHA-256 hash — same everywhere
Data location	Fixed — us-east-1 S3	Dynamic — nearest to compute
De-duplication	Manual, expensive	Automatic — same hash = one copy
Data movement	Human ETL job	Gravity algorithm — autonomous
Verification	Trust the storage provider	Re-hash and compare — trustless
Garbage collect	Manual lifecycle policies	Automatic — cold copies removed

4. Conscious Compliance Engine

DPDP 2023, GDPR, HIPAA, and SOX are encoded as executable Python expressions on Hyperledger Fabric smart contracts. Every data transfer is evaluated against applicable laws before execution. Auto-fixes resolve ~60% of violations.

Law	Rule	Check	Auto-fix
DPDP	DPDP-001	No Indian PII outside IN without consent	request_consent()
DPDP	DPDP-002	Transfer purpose must match consent purpose	None — human review
DPDP	DPDP-003	All processing logged to Hyperledger Fabric	log_to_fabric()
GDPR	GDPR-001	EU data stays in adequate countries only	None — human needed
GDPR	GDPR-002	Data must be erasable on request	mark_for_erasure()
GDPR	GDPR-003	Data minimisation — only necessary fields	drop_extra_cols()
HIPAA	HIPAA-001	PHI encrypted at rest and in transit	encrypt_now()
HIPAA	HIPAA-002	Access requires minimum necessary clearance	None — deny

5. AI Operating System

The AI OS receives human intentions, classifies them, generates Airflow DAGs and Spark jobs, deploys them, monitors sigma levels, and heals failures. 80%+ of pipeline failures resolved without human intervention.

Failure Type	Diagnosis	Action	Rate
connection_timeout	Source timed out	Retry exponential backoff x3	95%
schema_mismatch	New column in source	Quarantine + schema evolution	80%
quality_drop	Sigma below 4.5	Quarantine + root-cause ML	70%
oom_error	Executor out of memory	Reduce batch + increase memory	90%
source_unavailable	Source system down	Switch to backup source	85%
unknown_error	Unrecognised	Escalate to human operator	0%

6. REST API Reference

Method	Endpoint	Description
GET	/health	Platform health, sigma avg, node count
POST	/api/v1/ingest	Ingest data — creates DNA, enters fabric, logs to Fabric
GET	/api/v1/dataset/{id}	Retrieve dataset with DNA check and access control
POST	/api/v1/compliance/border	Check transfer legality against DPDP/GDPR/HIPAA
GET	/api/v1/compliance/report/{id}	Generate 60-second blockchain-verified compliance report
POST	/api/v1/query/nlp	NLP query in Telugu/Hindi/English to Presto SQL
GET	/api/v1/fabric/status	Zero Gravity fabric: nodes, datasets, movements
POST	/api/v1/pipeline/intent	Natural language intent to AI OS pipeline builder
GET	/api/v1/sigma/{pipeline_id}	Live Six Sigma score for any pipeline
POST	/api/v1/node/register	Register new living data node in the fabric

POST /api/v1/ingest

```
{
  "node_id": "dn-hyderabad-01",
  "name": "patient_jan25",
  "creator_id": "apollo_hyd",
  "sigma": 5.8,
  "laws": ["IN_DPDP_2023"],
  "allowed_regions": ["IN", "IN-TG"],
  "purpose": "medical_treatment",
  "data": "patient_id,age\nP001,45"
}
```

200 OK: {"status": "ingested", "genome_hash": "sha256-...", "sigma": 5.8, "fabric_nodes": 2}

POST /api/v1/query/nlp

```
{
  "text": "■■■■■ ■■■■■ ■■■■■■■■■",
  "language": "te",
  "tables": ["sales"]
}
```

200 OK: {"sql": "SELECT region, SUM(revenue) FROM sales WHERE ...", "engine": "presto"}

7. Deployment

```
# Full stack in 3 commands
git clone https://github.com/datanexus/core.git && cd core
docker-compose -f era3/docker-compose.yml up -d
curl http://localhost:8000/health

# Run all 5 Era 3 demos
python era3/dna/data_dna.py          # Data DNA
python era3/nodes/living_node.py     # Living Nodes
python era3/fabric/zero_gravity.py   # Zero Gravity
python era3/compliance/conscious_compliance.py # Conscious Compliance
python era3/api/api.py               # REST API demo
```

Mode	Infrastructure	Best For	DPDP Compliant
Cloud-native	AWS EKS / GCP GKE	Startups, scale-ups	Partial
Hybrid	Cloud + on-premise	Banks, hospitals	Yes
On-premises	Customer machines	Governments, defence	Yes — full
Air-gap	Zero internet	Military, critical infra	Yes — maximum
Multi-cloud	AWS + GCP + Azure together	Multinationals	Yes — federated

8. What Was Built

Component	Status	Language	License
Data DNA	COMPLETE	Python	Apache 2.0
Living Data Nodes	COMPLETE	Python	Apache 2.0
Zero Gravity Fabric	COMPLETE	Python	Apache 2.0
Conscious Compliance	COMPLETE	Python	Apache 2.0
AI Operating System	COMPLETE	Python	Apache 2.0
REST API Gateway	COMPLETE	Python	Apache 2.0
Docker Compose Stack	COMPLETE	YAML	Apache 2.0
GitHub README	COMPLETE	Markdown	Apache 2.0
Architecture PDF	COMPLETE	Document	Apache 2.0
Investor Pitch PPTX	COMPLETE	10 slides	Apache 2.0
Manifest Document	COMPLETE	Document	Apache 2.0
90-Day Launch Plan	COMPLETE	Document	Apache 2.0

*"Karmanye vadhikaraste ma phaleshu kadachana" — You have the right to perform your duty.
Never to the fruits thereof.*

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Open Source · Dharma-driven · Six Sigma · Blockchain-verified